

Ordinary Differential Equations

A First-Principles Cheat Sheet for Higher-Order Methods, Variable Coefficients & Phase-Plane Systems

Color key: all live solution steps in the **Worked Examples** are typeset in purple; explanatory text stays black. This guide picks up where the constant-coefficient second-order story ends—it asks how to manufacture a missing solution, how to climb past order two, how to force *any* right-hand side, what happens when the coefficients themselves depend on the variable, and finally how a system of equations draws its own geometry in the phase plane. Five tools, one decision map at the end.

Notation at a glance

$y'' + p(t)y' + q(t)y = g(t)$	standard form (leading coefficient = 1).
$W(y_1, \dots, y_n)$	Wronskian determinant—measures linear independence.
$y_1, y_2, \dots$	a <i>fundamental set</i> : n independent solutions of the homogeneous ODE.
y_c, y_p	complementary (homogeneous) and particular solutions; $y = y_c + y_p$.
$x = e^t, y = x^r$	substitutions that linearize a Cauchy–Euler equation.
$J = \partial(f, g)/\partial(x, y)$	Jacobian; $\tau = \text{tr } J$, $\Delta = \det J$ classify equilibria.

1. Reduction of Order

The Core Logic

A second-order linear homogeneous ODE has a *two-dimensional* solution space—it is not solved until two linearly independent solutions are in hand. Reduction of order answers the question: given one solution $y_1(t)$, how do we manufacture a second, independent $y_2(t)$? The device is to look for $y_2 = v(t)y_1(t)$ —a **modulation** of the known solution by an unknown function v . If v were constant, y_2 would just be a scalar multiple of y_1 and contribute nothing new; so any genuinely independent solution must let v *vary*.

Work in standard form $y'' + p(t)y' + q(t)y = 0$. Substituting $y_2 = vy_1$:

$$y_2' = v'y_1 + vy_1', \quad y_2'' = v''y_1 + 2v'y_1' + vy_1''.$$

Insert into the ODE and regroup by derivatives of v :

$$y_1 v'' + (2y_1' + p y_1)v' + \underbrace{(y_1'' + p y_1' + q y_1)}_{=0 \text{ (} y_1 \text{ solves the ODE)}} v = 0.$$

The v -term vanishes identically—*this is the whole trick*. What remains is first order in $w = v'$:

$$y_1 w' + (2y_1' + p y_1) w = 0 \implies \frac{w'}{w} = -\frac{2y_1'}{y_1} - p.$$

Integrate, exponentiate, and back-substitute $v = \int w dt$ to obtain the **reduction-of-order formula**:

$$w = \frac{e^{-\int p dt}}{y_1^2}, \quad \boxed{y_2 = y_1 \int \frac{e^{-\int p dt}}{y_1^2} dt}.$$

Independence is automatic. With $y_2 = v y_1$ the Wronskian collapses to $W(y_1, y_2) = y_1 y_2' - y_2 y_1' = v' y_1^2 = e^{-\int p dt} \neq 0$ —this is Abel's formula, and an exponential never vanishes, so y_1, y_2 are independent wherever the ODE is defined.

All Possible Cases / Variations

- **No first-derivative term** ($p = 0$): the formula simplifies to $y_2 = y_1 \int dt / y_1^2$.
- **Constant-coefficient repeated root.** $y'' - 2r_1 y' + r_1^2 y = 0$ with $y_1 = e^{r_1 t}$ yields $y_2 = t e^{r_1 t}$ —this is *why* a double root spawns the extra factor of t .
- **Cauchy–Euler repeated root.** $y_1 = x^r$ produces $y_2 = x^r \ln x$ (Section 4).
- **Constants of integration.** Any $+C$ in $v = \int w dt$ merely re-adds a multiple of y_1 —drop it.
- **Variable coefficients.** The method's headline strength: it needs no characteristic equation, only one known solution and a (possibly hard) integral.

Step-by-Step Methodology

1. Divide by the leading coefficient to reach standard form; read off $p(t)$.
2. Identify the known solution y_1 .
3. Compute $\mu = e^{-\int p dt}$.
4. Form $w = \mu / y_1^2$.
5. Integrate $v = \int w dt$ (discard the constant).
6. Assemble $y_2 = v y_1$; the Wronskian is guaranteed nonzero.
7. General solution $y = c_1 y_1 + c_2 y_2$.

Worked Example A — Variable Coefficients

$t^2 y'' - 3t y' + 4y = 0$ on $t > 0$, given $y_1 = t^2$.

Step 1 — Standard form *divide by t^2* :

$$y'' - \frac{3}{t} y' + \frac{4}{t^2} y = 0, \quad p(t) = -\frac{3}{t}.$$

Step 2 — Integrating factor $\mu = e^{-\int p dt}$:

$$\int p dt = \int -\frac{3}{t} dt = -3 \ln t \implies \mu = e^{3 \ln t} = t^3.$$

Step 3 — Build $w = \mu / y_1^2$:

$$w = \frac{t^3}{(t^2)^2} = \frac{t^3}{t^4} = \frac{1}{t}.$$

Step 4 — Integrate for v :

$$v = \int \frac{1}{t} dt = \ln t.$$

Step 5 — Second solution and general solution:

$$y_2 = v y_1 = t^2 \ln t, \quad y = c_1 t^2 + c_2 t^2 \ln t.$$

Check: $W = t^2(2t \ln t + t) - t^2 \ln t(2t) = t^3 \neq 0$ for $t > 0$. Independent.

Worked Example B — Recovering the Repeated-Root Factor of t

$y'' - 6y' + 9y = 0$, given $y_1 = e^{3t}$.

Step 1 — Integrating factor *here* $p = -6$:

$$\int p \, dt = -6t \implies \mu = e^{6t}.$$

Step 2 — Build w :

$$w = \frac{e^{6t}}{(e^{3t})^2} = \frac{e^{6t}}{e^{6t}} = 1.$$

Step 3 — Integrate and assemble:

$$v = \int 1 \, dt = t, \quad y_2 = t e^{3t}.$$

Result: $y = (c_1 + c_2 t) e^{3t}$ —the missing $t e^{3t}$ derived from first principles.

2. Linear Higher-Order Equations

The Core Logic

An n -th order linear homogeneous ODE with constant coefficients,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0,$$

has an n -dimensional solution space. The ansatz $y = e^{rt}$ turns every derivative into a power of r (since $y^{(k)} = r^k e^{rt}$); substituting and cancelling $e^{rt} \neq 0$ leaves the **characteristic (auxiliary) polynomial**

$$a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = 0.$$

By the Fundamental Theorem of Algebra this has exactly n roots counted with multiplicity, and each root—real or complex, simple or repeated—contributes basis solutions. A **fundamental set** is any n linearly independent solutions; the general solution is their span.

Linear independence via the Wronskian.

$$W(y_1, \dots, y_n)(t) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}.$$

For solutions of a linear ODE, **Abel's theorem** gives $W(t) = W(t_0) \exp\left(-\int \frac{a_{n-1}}{a_n} dt\right)$. An exponential is never zero, so W is *either identically zero or never zero*—checking one point decides independence everywhere.

All Possible Cases / Variations

- **Distinct real root** r : contributes e^{rt} .
- **Real root of multiplicity** m : $e^{rt}, te^{rt}, \dots, t^{m-1}e^{rt}$ (m functions).
- **Simple complex pair** $\alpha \pm i\beta$: $e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t$.
- **Complex pair of multiplicity** m : $t^j e^{\alpha t} \cos \beta t$ and $t^j e^{\alpha t} \sin \beta t$ for $j = 0, \dots, m-1$ ($2m$ functions).
- The basis-function count always sums to exactly n —a built-in sanity check.

Root type	Basis contribution
real, simple r	e^{rt}
real, multiplicity m	$e^{rt}, te^{rt}, \dots, t^{m-1}e^{rt}$
complex $\alpha \pm i\beta$, simple	$e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t$
complex, multiplicity m	$t^j e^{\alpha t} \cos \beta t, t^j e^{\alpha t} \sin \beta t, j = 0, \dots, m-1$

Step-by-Step Methodology

1. Write the characteristic polynomial.
2. Factor it; find all n roots with multiplicities (rational-root test, synthetic division, known forms).
3. Assign basis functions from the case table.
4. General solution = linear combination of all basis functions.
5. (IVP) Apply $y(t_0), y'(t_0), \dots, y^{(n-1)}(t_0)$; solve the linear system. Nonzero W guarantees a unique solution.

Worked Example A — Triple Real Root

$$y''' - 3y'' + 3y' - y = 0.$$

Step 1 — Characteristic equation *recognize the binomial cube:*

$$r^3 - 3r^2 + 3r - 1 = (r - 1)^3 = 0 \implies r = 1 \text{ (multiplicity 3).}$$

Step 2 — Basis from a root of multiplicity 3:

$$e^t, te^t, t^2e^t.$$

Result: $y = (c_1 + c_2t + c_3t^2)e^t$.

Worked Example B — Mixed Real and Complex Roots

$$y^{(4)} - 16y = 0.$$

Step 1 — Factor the characteristic polynomial *difference of squares, twice:*

$$r^4 - 16 = (r^2 - 4)(r^2 + 4) = (r - 2)(r + 2)(r^2 + 4) = 0.$$

Step 2 — Read off all four roots:

$$r = 2, -2, +2i, -2i.$$

Step 3 — Assign basis functions:

$$e^{2t}, \quad e^{-2t}, \quad \cos 2t, \quad \sin 2t.$$

Result: $y = c_1 e^{2t} + c_2 e^{-2t} + c_3 \cos 2t + c_4 \sin 2t.$

3. Variation of Parameters

The Core Logic

For the non-homogeneous equation $y'' + py' + qy = g(t)$ with a known fundamental set $\{y_1, y_2\}$ of the homogeneous problem, *undetermined coefficients* only works when the coefficients are constant *and* g is a polynomial, exponential, sine/cosine, or product of these. **Variation of parameters removes both restrictions**—it builds a particular solution for *any* continuous g and *any* fundamental set, including variable-coefficient problems. The idea: promote the constants in $y_c = c_1 y_1 + c_2 y_2$ to *functions*,

$$y_p = u_1(t) y_1 + u_2(t) y_2.$$

Derivation. Differentiate: $y'_p = u'_1 y_1 + u'_2 y_2 + u_1 y'_1 + u_2 y'_2$. Two unknown functions, one equation—so we are free to *impose one constraint*. Choose

$$u'_1 y_1 + u'_2 y_2 = 0. \quad (*)$$

Then $y'_p = u_1 y'_1 + u_2 y'_2$ and $y''_p = u'_1 y'_1 + u'_2 y'_2 + u_1 y''_1 + u_2 y''_2$. Substitute into the ODE and group:

$$u_1 \underbrace{(y''_1 + p y'_1 + q y_1)}_{=0} + u_2 \underbrace{(y''_2 + p y'_2 + q y_2)}_{=0} + u'_1 y'_1 + u'_2 y'_2 = g,$$

so

$$u'_1 y'_1 + u'_2 y'_2 = g. \quad (**)$$

Equations (*) and (**) form a 2×2 linear system whose coefficient matrix is the Wronskian. **Cramer's rule** delivers

$$u'_1 = \frac{-y_2 g}{W}, \quad u'_2 = \frac{y_1 g}{W}, \quad W = y_1 y'_2 - y_2 y'_1.$$

Integrate and assemble the closed-form particular solution:

$$y_p = -y_1 \int \frac{y_2 g}{W} dt + y_2 \int \frac{y_1 g}{W} dt.$$

Order n . The same construction with Cramer's rule gives $u'_k = g(t) W_k(t)/W(t)$, where W_k is W with its k -th column replaced by $(0, \dots, 0, 1)^\top$, and g is the right-hand side *after* normalizing the leading coefficient to 1.

All Possible Cases / Variations

- **Constant coefficients, “nice” g :** undetermined coefficients is usually faster—use it.
- **Variable coefficients:** find the homogeneous set first (Cauchy–Euler, reduction of order), then VoP.
- **Forcing UC cannot handle** ($g = \sec t, \tan t, \ln t, 1/t, \dots$): VoP is mandatory.
- **Standard form is essential**— g must be the RHS *after* dividing out the leading coefficient.
- **Constants of integration** in u_1, u_2 just rebuild y_c ; drop them.

Feature	Undetermined Coefficients	Variation of Parameters
Coefficients	constant only	constant <i>or</i> variable
Forcing $g(t)$	poly, exp, sin/cos & products	any continuous g
Mechanism	guess a trial, match	integrate the Wronskian formula
Main cost	algebra	integration (can be hard)
Fails when	$g = \tan t, \sec t, \ln t$	integrals are intractable

Step-by-Step Methodology

1. Standard form (leading coefficient 1); identify $g(t)$.
2. Solve the homogeneous equation for $\{y_1, y_2\}$.
3. Compute $W = y_1 y_2' - y_2 y_1'$.
4. Form $u_1' = -y_2 g / W$ and $u_2' = y_1 g / W$.
5. Integrate for u_1, u_2 (drop constants).
6. $y_p = u_1 y_1 + u_2 y_2$; simplify.
7. General solution $y = c_1 y_1 + c_2 y_2 + y_p$.

Worked Example — Forcing That Defeats Undetermined Coefficients

$$y'' + y = \sec t.$$

Step 1 — Homogeneous set $\sec t$ has no finite trial—UC is impossible:

$$r^2 + 1 = 0 \Rightarrow r = \pm i, \quad y_1 = \cos t, \quad y_2 = \sin t.$$

Step 2 — Wronskian:

$$W = \cos t \cos t - \sin t(-\sin t) = \cos^2 t + \sin^2 t = 1.$$

Step 3 — Derivatives of the parameters:

$$u_1' = \frac{-y_2 g}{W} = -\sin t \sec t = -\tan t, \quad u_2' = \frac{y_1 g}{W} = \cos t \sec t = 1.$$

Step 4 — Integrate:

$$u_1 = \int -\tan t \, dt = \ln |\cos t|, \quad u_2 = \int 1 \, dt = t.$$

Step 5 — Particular solution:

$$y_p = u_1 y_1 + u_2 y_2 = \cos t \ln |\cos t| + t \sin t.$$

Result: $y = c_1 \cos t + c_2 \sin t + \cos t \ln |\cos t| + t \sin t.$

4. Variable-Coefficient Equations: Cauchy–Euler

The Core Logic

A **Cauchy–Euler (equidimensional)** equation pairs each derivative with a matching power of the independent variable:

$$ax^2y'' + bxy' + cy = 0, \quad x > 0.$$

The structure is *scale-invariant* (unchanged under $x \mapsto \lambda x$), which is exactly why it admits power-law solutions $y = x^r$. Substitute $y = x^r$, $y' = rx^{r-1}$, $y'' = r(r-1)x^{r-2}$:

$$ax^2r(r-1)x^{r-2} + bxx^r + cx^r = x^r[ar(r-1) + br + c] = 0.$$

Cancel $x^r \neq 0$ to reach the **indicial equation**

$$ar(r-1) + br + c = 0 \iff ar^2 + (b-a)r + c = 0.$$

Equivalent view. The substitution $x = e^t$ (so $t = \ln x$) sends $\frac{d}{dx} = \frac{1}{x} \frac{d}{dt}$ and $\frac{d^2}{dx^2} = \frac{1}{x^2} \left(\frac{d^2}{dt^2} - \frac{d}{dt} \right)$, turning the equation into the *constant-coefficient* ODE $ay'' + (b-a)y' + cy = 0$ —whose characteristic equation is the indicial equation. The power-law ansatz and the log-substitution are two faces of one idea.

All Possible Cases / Variations

- **Distinct real roots** $r_1 \neq r_2$: $y = c_1 x^{r_1} + c_2 x^{r_2}$.
- **Repeated root** r : $y = (c_1 + c_2 \ln x)x^r$ —the $\ln x$ is the reduction-of-order partner of x^r (mirroring te^{rt}).
- **Complex roots** $r = \lambda \pm i\mu$: $y = x^\lambda [c_1 \cos(\mu \ln x) + c_2 \sin(\mu \ln x)]$, since $x^{\lambda \pm i\mu} = x^\lambda e^{\pm i\mu \ln x}$.
- **Domain** $x < 0$: replace x by $|x|$ throughout.
- **Higher order**: same ansatz $y = x^r$; the indicial polynomial has degree n .
- **Non-homogeneous**: substitute $x = e^t$ then use undetermined coefficients in t , or apply VoP directly in x .

Step-by-Step Methodology

1. Confirm equidimensional form $ax^2y'' + bxy' + cy = 0$.
2. Write the indicial equation $ar^2 + (b-a)r + c = 0$ (or substitute $y = x^r$ directly).
3. Solve for r ; classify the roots.
4. Write the matching solution form (real / repeated / complex).
5. (If forced) convert via $x = e^t$ or use variation of parameters.

Worked Example A — Distinct Real Roots

$$x^2y'' + 2xy' - 6y = 0.$$

Step 1 — Indicial equation $a = 1, b = 2, c = -6$:

$$r^2 + (2 - 1)r - 6 = r^2 + r - 6 = (r + 3)(r - 2) = 0.$$

Step 2 — Roots and solution:

$$r = 2, -3 \implies y = c_1 x^2 + c_2 x^{-3}.$$

Worked Example B — Repeated Root

$$x^2 y'' - 3xy' + 4y = 0.$$

Step 1 — Indicial equation $a = 1, b = -3, c = 4$:

$$r^2 + (-3 - 1)r + 4 = r^2 - 4r + 4 = (r - 2)^2 = 0.$$

Step 2 — Double root forces a $\ln x$:

$$r = 2 \text{ (double)} \implies y = (c_1 + c_2 \ln x) x^2.$$

Worked Example C — Complex Roots

$$x^2 y'' + xy' + y = 0.$$

Step 1 — Indicial equation $a = 1, b = 1, c = 1$:

$$r^2 + (1 - 1)r + 1 = r^2 + 1 = 0 \implies r = \pm i, \quad \lambda = 0, \mu = 1.$$

Step 2 — Real-valued solution:

$$y = x^0 [c_1 \cos(\ln x) + c_2 \sin(\ln x)] = c_1 \cos(\ln x) + c_2 \sin(\ln x).$$

5. Phase-Plane Analysis & Phase Diagrams

The Core Logic

A planar **autonomous system**

$$x' = f(x, y), \quad y' = g(x, y)$$

carries no explicit t on the right—its entire behavior lives as geometry in the xy *phase plane*, where solutions trace trajectories (orbits) everywhere tangent to the vector field (f, g) . **Critical (equilibrium) points** solve $f = g = 0$: the system is at rest there. Near a critical point (x_*, y_*) the nonlinear field is well approximated by its **linearization**, the Jacobian

$$J = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} \bigg|_{(x_*, y_*)}.$$

Its eigenvalues govern the local flow. Writing $\tau = \text{tr } J$ and $\Delta = \det J$,

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}.$$

Real eigenvalues stretch or compress along eigenvector directions (straight-line orbits); *complex* eigenvalues rotate (spirals or centers). The *sign of the real part* sets stability. A second-order ODE $x'' + px' + qx = 0$ becomes such a system via $x_1 = x$, $x_2 = x'$ —so the damping regimes are these portraits in disguise: **overdamped** = **node**, **underdamped** = **spiral**, **undamped** = **center**.

All Possible Cases / Variations

- **Linear system** $\mathbf{x}' = A\mathbf{x}$: $J = A$ everywhere; a single critical point at the origin when $\det A \neq 0$.
- **Eigenvector geometry**: real eigenvalues' eigenvectors *are* the straight-line trajectories—fast/slow axes of a node, separatrices of a saddle.
- **Borderline cases** ($\tau = 0$ center, or a zero eigenvalue): linearization is *inconclusive* for the nonlinear system—a center may in truth be a slow spiral.
- **Rotation sense** of a spiral/center: read it from the field at one test point.
- **Hartman–Grobman**: linearization faithfully captures the nonlinear portrait whenever every eigenvalue has nonzero real part (a *hyperbolic* equilibrium).

Eigenvalues	Type	Stability
real, distinct, same sign	node	stable if both < 0 ; unstable if both > 0
real, opposite signs	saddle	unstable (always)
complex, $\text{Re} \neq 0$	spiral (focus)	stable if $\text{Re} < 0$; unstable if > 0
pure imaginary	center	neutrally stable (closed orbits)
repeated real	star / improper node	sign of the repeated root

Trace–determinant quick read.

- $\Delta < 0 \Rightarrow$ saddle.
- $\Delta > 0$, $\tau^2 - 4\Delta > 0 \Rightarrow$ node ($\tau < 0$ stable).
- $\Delta > 0$, $\tau^2 - 4\Delta < 0 \Rightarrow$ spiral ($\tau < 0$ stable).
- $\tau = 0$, $\Delta > 0 \Rightarrow$ center.

Step-by-Step Methodology

1. Set $f = 0$, $g = 0$; solve for *all* critical points.
2. Compute the Jacobian $J(x, y)$ symbolically.
3. Evaluate J at each critical point.
4. Find the eigenvalues—or just τ, Δ .
5. Classify type and stability.
6. (Optional) eigenvectors give straight-line directions; sketch orbits consistent with the field.

Worked Example A — A Linear Saddle

$$x' = x + 2y, \quad y' = 3x + 2y.$$

Step 1 — Jacobian (constant for a linear system) $\det A \neq 0$, so the origin is the only equilibrium:

$$J = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}, \quad \tau = 3, \quad \Delta = (1)(2) - (2)(3) = -4.$$

Step 2 — Classify by Δ :

$$\Delta = -4 < 0 \implies \text{saddle (unstable)}.$$

Step 3 — Confirm with eigenvalues:

$$\lambda = \frac{3 \pm \sqrt{9+16}}{2} = \frac{3 \pm 5}{2} = 4, -1 \quad (\text{opposite signs}).$$

Worked Example B — A Nonlinear System (Damped Pendulum)

$$x' = y, \quad y' = -\sin x - y.$$

Step 1 — Critical points $y = 0$ and $\sin x = 0 \Rightarrow x = n\pi$:

$$(0, 0) \quad \text{and} \quad (\pi, 0) \quad (\text{representative of all } n\pi).$$

Step 2 — Jacobian:

$$J = \begin{pmatrix} 0 & 1 \\ -\cos x & -1 \end{pmatrix}.$$

Step 3 — Hanging rest position $(0, 0)$:

$$J = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, \quad \tau = -1, \quad \Delta = 1, \quad \tau^2 - 4\Delta = -3 < 0 \implies \text{stable spiral}.$$

Step 4 — Inverted position $(\pi, 0)$:

$$J = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}, \quad \tau = -1, \quad \Delta = -1 < 0 \implies \text{saddle (unstable)}.$$

Reading: the damped pendulum spirals into rest at the bottom and is a saddle when balanced upright—exactly the physical picture.

6. Quick-Reference Diagnostic Map

Given an arbitrary equation, reduce to standard form, then ask—in order:

1. **Is it a system** $x' = f$, $y' = g$ (or is a phase portrait requested)? $\rightarrow$ **Phase plane:** critical points, Jacobian, eigenvalues. (§5)
2. **Single linear ODE—homogeneous or forced?**

3. Constant coefficients?

- Homogeneous, constant coeff, order $n \rightarrow$ **characteristic equation** (§2).
- One solution y_1 already known, cannot factor $\rightarrow$ **reduction of order** (§1).
- Equidimensional $ax^2y'' + bxy' + cy \rightarrow$ **Cauchy–Euler** (§4).

4. Non-homogeneous $y'' + py' + qy = g$:

- Constant coeff *and* g is poly/exp/sin-cos $\rightarrow$ **undetermined coefficients** (fastest).
- Variable coeff, *or* $g = \sec t, \tan t, \ln t, \dots \rightarrow$ **variation of parameters** (§3).

Equation you see	Diagnostic	Tool
$ay'' + by' + cy = 0$	const coeff, homogeneous	characteristic eqn (§2)
order ≥ 3 , const coeff	factor degree- n poly	characteristic eqn (§2)
one solution y_1 known	need a 2nd independent	reduction of order (§1)
$ax^2y'' + bxy' + cy = 0$	powers match derivatives	Cauchy–Euler (§4)
$y'' + py' + qy = g$, nice g , const	match a trial form	undetermined coeff
$y'' + py' + qy = g$, $g = \sec, \tan, \ln$	UC has no trial	variation of params (§3)
$x' = f(x, y)$, $y' = g(x, y)$	autonomous system	phase plane (§5)
$x'' + px' + qx = 0$ via (x, x')	recast as a system	phase plane (§5)

Many problems chain these tools—a forced Cauchy–Euler equation needs §4 for its homogeneous set and then §3 for the particular solution; a higher-order forced ODE needs §2 then variation of parameters. **Always normalize to standard form first**, then read three forks: constant vs. variable coefficients, homogeneous vs. forced, single equation vs. system. Those three answers route every problem to exactly one of the five methods above.